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If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as necessary.

Clean-up
After discharge of this cargo, the bilge wells and the scuppers of the cargo spaces shall be checked and any blockage in the bilge wells and the scuppers shall be removed."

"AMMONIUM NITRATE BASED FERTILIZER MHB

Ammonium nitrate based fertilizers transported under conditions mentioned in this schedule are uniform mixtures of nitrogen with or without potash and/or phosphate within th e following composition limits:

.1 not more than 70% ammonium nitrate and not more than 0.4% total combustible organic material calculated as carbon or not more than 45% ammonium nitrate and unrestricted combustible material; and

.2 both the ammonium nitrate content is equal to or greater than 20% and the chloride content is equal to or greater than 2% (as indicated in the grey area of the figure below).

SOURCE: MSC.500(105).pdf
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chloride content is equal to or greater than 2% (as indicated in the grey area of the figure below).

The shipper shall declare the ammonium nitrate content and the chloride content in accordance with 4.2 of this Code.

Notwithstanding the above, fertilizers within these composition limits are not subject to the provisions of this schedule, if they are assigned class 9 due to the hazard of self -sustaining decomposition based on the results o f the trough test (referred to in the UN Manual of Tests and Criteria, part III, section 39).

% Ammonium
nitrate
% Chloride
70 %
20 %
2 %
RESOLUTION MSC.500(105) (adopted on 28 April 2022)
AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE

Description
Crystals, granules or prills. Non -cohesive when dry. Wholly or partly soluble in water.
Common products listed under this schedule are compound NPK/NK fertilizers.

Characteristics
Physical properties

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Common products listed under this schedule are compound NPK/NK fertilizers.

Characteristics
Physical properties
Size Angle of repose Bulk density (kg/m3) Stowage factor (m3/t)
1 to 5 mm 27° to 42° 1,000 to 1,200 0.83 to 1.00
Hazard classification
Class Subsidiary hazard(s) MHB Group
Not applicable Not applicable OH B

Hazard
Although this cargo passes the trough test (referred to in the UN Manual of Tests and Criteria, part III, section 39), and hence does not fall in class 9, when carried in bulk in large quantities, it may still be subject to decomposition if strongly heated from external sources. Once initiated, decomposition might gradually spread through the remainder of the cargo, producing large volumes of toxic gases.

This cargo is not subject to an explosion hazard.

Fertilizer dust might be irritating to skin and mucous membranes. It is hygroscopic cargo and will cake if wet.

Stowage and segregation

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Fertilizer dust might be irritating to skin and mucous membranes. It is hygroscopic cargo and will cake if wet.

Stowage and segregation
Separated from sources of heat (see also Loading). Not to be stowed immediately adjacent to any tank, double bottom or pipe containing heated fuel oil, unless there are permanent means and procedures to monitor and control the temperature so that it does not exceed 50°C. Fertilizers of this type shall be stowed out of direct contact with a metal engine-room boundary. This may be done, for example, by using flame-retardant bags containing inert materials or by any equivalent arrangement approved by the competent authority of the country of origin. This requirement does not apply if the bulkhead is class A-60 or to short international voyages.

The hatches of the cargo spaces, including those of 'tween decks, shall be kept free at all times. In case of an emergency, whenever this material is on board, opening the hatches must
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times. In case of an emergency, whenever this material is on board, opening the hatches must be enabled (see 9.3.1.13 of this Code).

Hold cleanliness
Clean and dry as relevant to the hazards of the cargo.

Weather precautions
This cargo shall be kept as dry as practicable. This cargo shall not be handled during precipitation. During handling of this cargo, all non -working hatches of the cargo spaces into which this cargo is loaded, or is to be loaded, shall be closed.

Loading
Trim in accordance with the relevant provisions required under sections 4 and 5 of this Code.

Prior to loading, the following provisions shall be complied with
In order to avoid heating up of the cargo, all electrical equipment or other equipment capable of developing heat, other than that of approved safe type, in the cargo spaces to be used for this cargo, shall be electrically disconnected from the power source, by appropriate means
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this cargo, shall be electrically disconnected from the power source, by appropriate means other than a fuse, at a point external to the space. This situation shall be maintained while the cargo is on board.

Due consideration shall be given to the necessity to open hatches in case of decomposition to provide maximum ventilation, release pressure and heat, and slow down the reaction.

During loading, the following provisions shall be complied with
Organic contamination aggravates decomposition behaviour in the presence of sources of heat, and therefore bunkering of fuel oil shall not be allowed during loading.

Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed.

Precautions
NO SMOKING signs shall be displayed on deck whenever this cargo is on board. Smoking
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engine-room, shall not be allowed.

Precautions
NO SMOKING signs shall be displayed on deck whenever this cargo is on board. Smoking shall not be allowed on deck and in the cargo spaces.
No welding, burning, cutting or other operations involving the use of fire, open flame, spark or arc-producing equipment shall be carried out on equipment or structures in direct contact with

the fertilizer.

The master and officers are to note that the ship's fixed gas fire -fighting installation will be ineffective on decompositions involving this cargo and must not be used. If decomposition is identified, water must be applied without delay. Injection to the seat of decomposition is the first control measure because it uses less water and can be more effective in early decomposition stages. Total flooding is the final control measure but can introduce stability and stress issues. The consequential risk to the stability of the ship through fluidization of the
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and stress issues. The consequential risk to the stability of the ship through fluidization of the
cargo must be taken into account in both cases. Application of water to the surface of the cargo is much less effective and can give a false sense of safety.

Appropriate precautions shall be taken to protect machinery and accommodation spaces from the dust of the cargo.

Bilge wells of the cargo spaces shall be protected from ingress of the cargo. Due consideration shall be given to protect equipment from the dust of the cargo. Persons who may be exposed to the dust of the cargo shall wear goggles or other equivalent dust eye-protection and dust filter masks. Those persons shall wear protective clothing, as necessary.

Ventilation
The cargo spaces carrying this cargo shall not be ventilated during the voyage, except in an emergency.

Carriage
There shall be a daily monitoring, recording and assessment of the trends of the cargo
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emergency.

Carriage
There shall be a daily monitoring, recording and assessment of the trends of the cargo temperature and oxygen concentration in the cargo space(s) throughout the voyage.